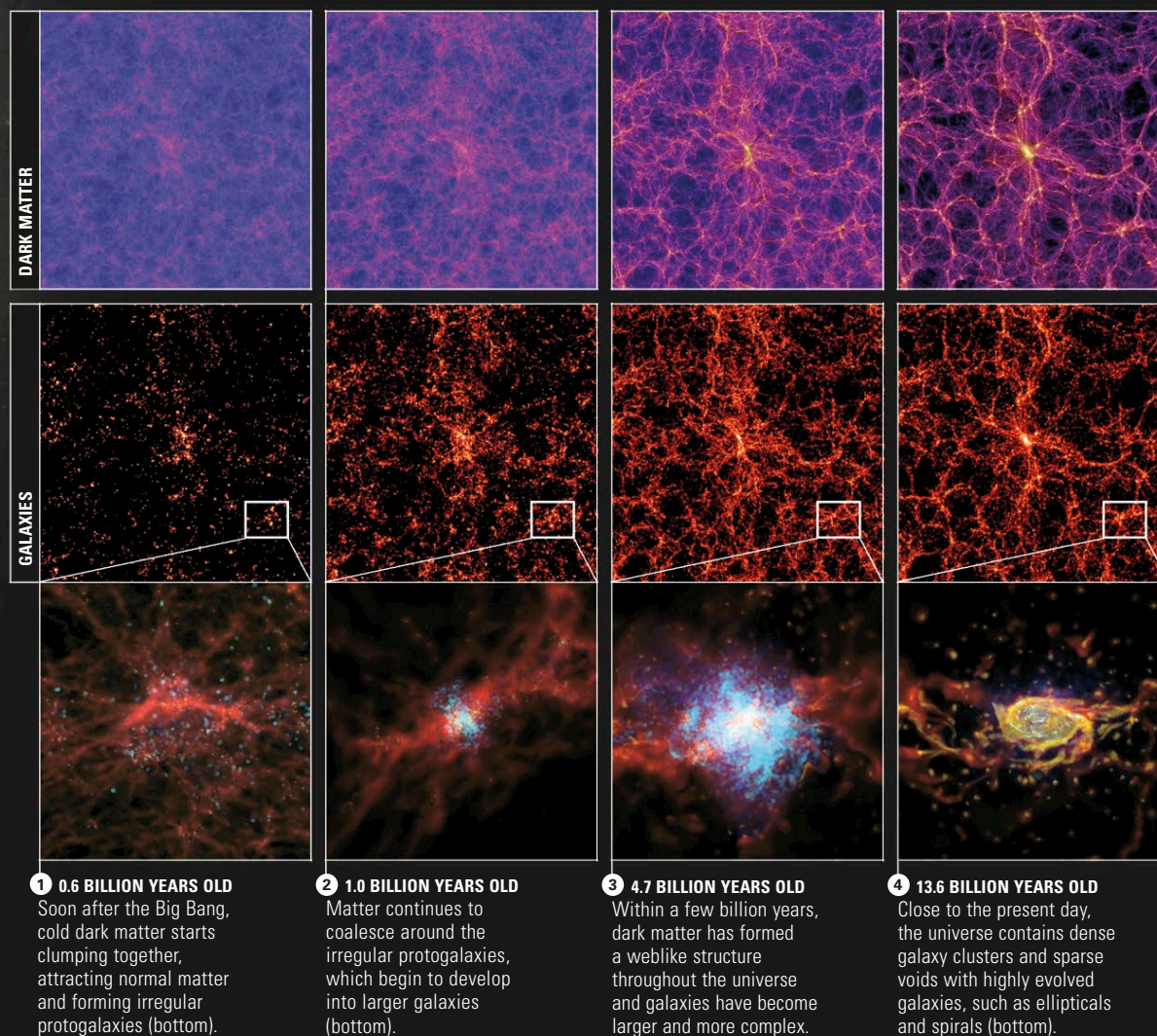


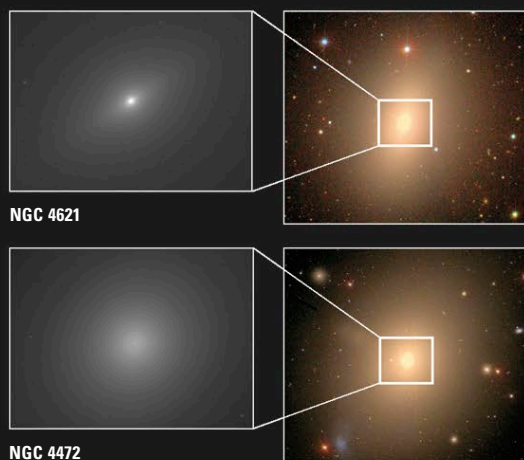
GALAXY FORMATION

Until recently, there were two main theories of galaxy formation. The first was a “top-down” scenario, in which galaxies coalesced out of huge clouds of matter, eventually becoming dense enough to form stars within them. The second was a “bottom-up” scenario in which small-scale structures formed first and gradually merged to create larger structures—galaxies. These two scenarios arose as a result of different ideas about the properties of dark matter (see p.27), specifically whether it is “hot” and fast-moving or “cold” and slow-moving. It now seems that the bottom-up model, with galaxy formation driven by the presence of relatively slow-moving cold dark matter (CDM), is correct. Computer simulations (such as the one shown below) suggest that in the early life of the universe, cold dark matter started to clump together in localized regions. These clumps acted as seeds, attracting yet more matter and eventually developing into protogalaxies and then mature galaxies. This happened in numerous localized regions throughout the universe, leading to the distribution of galaxies seen today.



BLACK HOLES IN GIANT ELLIPTICAL GALAXIES

A comparison of the giant elliptical galaxies NGC 4621 (top) and NGC 4472 (bottom) reveals few stars in the core of NGC 4472 compared to the bright center of NGC 4621. It is thought that this “star deficit” is due to many of NGC 4472’s stars being ejected during the violent collision of NGC 4472 with another galaxy and the merger of their central supermassive black holes.



THE ROLE OF BLACK HOLES

Since the 1990s, evidence has mounted to suggest that most if not all galaxies contain supermassive black holes at their centers, similar to the one at the center of our own Milky Way (see p.229). The masses of these black holes seem to be closely related to the overall sizes of the galaxies in which they lie, and a few galaxies even seem to contain two black holes in their cores. This suggests that black holes helped seed the formation of galaxies, and it also supports the “bottom-up” theory that larger galaxies are formed from mergers of smaller ones, with the central black holes ultimately joining together. However, the origin of these supermassive black holes is still unclear. Some theories suggest that the first black holes could have formed in the Big Bang or that they were created by the slow collapse of gas clouds around dark matter cores. Another possibility, which seems the most likely, is that they were formed by the death of an early generation of immense stars.